# The Reactivity Series

Aim: to observe displacement reactions and understand the reactivity series of metals.

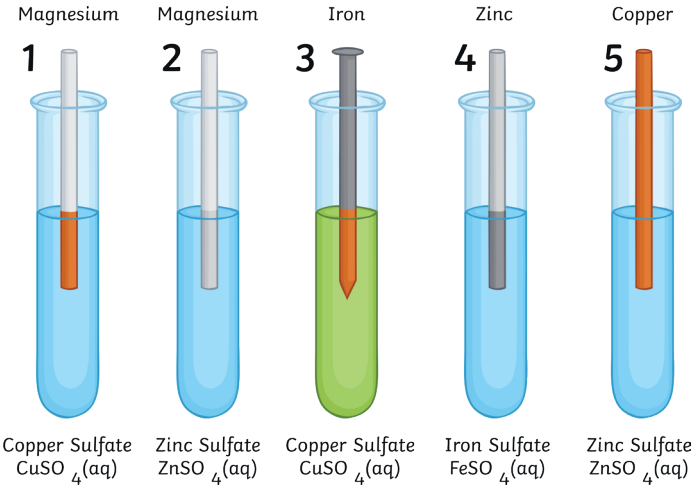
## Equipment:

- 5 test tubes
- Test tube rack
- Copper sulfate solution
- Zinc sulfate solution
- Iron sulfate solution
- Dropping pipettes
- Magnesium strips
- Copper strip
- Zinc strip
- Iron strip (or uncoated iron nail)
- Marker pen

## Safety Information

Glassware - risk of cuts if broken. Apply pressure to wounds and inform teacher immediately

Liquids - risk of slips and falls if spilt. Inform teacher and mop area immediately.




## Method:

Step 1: **You must wear goggles at all times**. Label the test tubes 1 to 5 and place them into the rack.

Step 2: Into test tube 1, place a piece of magnesium and half cover with copper sulfate solution. Observe any reactions and record your observation.

Step 3: Into test tube 2, place a piece of magnesium and half cover with zinc sulfate solution. Observe any reactions and record your observation.

Step 4: Into test tube 3, place a piece of iron and half cover with copper sulfate solution. Observe any reactions and record your observation.

Step 5: Into test tube 4, place a piece of zinc and half cover with iron sulfate solution. Observe any reactions and record your observation.

Step 6: Into test tube 5, place a piece of copper and half cover with zinc sulfate solution. Observe any reactions and record your observation.

NB: You may need to observe some reactions the next lesson. You could take photographs before and after to help compare and identify if a reaction occurred.



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## Results:

| Test tube | Observations | Reaction | Conclusion |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

## Questions:

1. Which metal was the most reactive?
2. Which metal was the least reactive?
3. How did you identify if a reaction was occurring?
4. What is the name of this type of reaction?
5. Complete the column of the table headed 'reaction' by writing a word equation for the reaction which you observed.
6. Complete the column of the table headed 'conclusion' by stating which metal is more reactive, e.g. metal X is more reactive than metal Y.



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# Answers

| Test tube | Observations | Reaction | Conclusion |
|---|---|---|---|
| 1 | Deposits of copper on the magnesium strip (orangey colour).<br>The solution turns from blue to colourless. | magnesium + copper sulfate → magnesium sulfate + copper | Magnesium is more reactive than copper. |
| 2 | Deposit of zinc on the magnesium strip (grey colour). | magnesium + zinc sulfate → magnesium sulfate + zinc | Magnesium is more reactive than zinc. |
| 3 | Deposits of copper on the iron strip (orangey colour).<br>The solution turns from blue to green. | iron + copper sulfate → iron sulfate + copper | Iron is more reactive than copper. |
| 4 | Deposits of iron on the zinc strip (dull grey colour). | zinc +iron sulfate → zinc sulfate + iron | Zinc is more reactive than iron. |
| 5 | No reaction. | | Zinc is more reactive than copper. |

1. Which metal was the most reactive?
   **Magnesium**
2. Which metal was the least reactive?
   **Copper**
3. How did you identify if a reaction was occurring?
   **Changes to the metal, changes to the solution.**
4. What is the name of this type of reaction?
   **Displacement reaction**
5. Complete the column of the table headed 'reaction' by writing a word equation for the reaction which you observed.
   **See above**
6. Complete the column of the table headed 'conclusion' by stating which metal is more reactive, e.g. metal X is more reactive than metal Y.
   **See above**